

Stable Diffusion: The Image Generation Wizard

Explore Stable Diffusion—an open source generative AI marvel that crafts images from text.



Generative AI centres on creating new and original content, text, images, synthetic data, or even deepfakes. It can generate output within the same interface as the input (e.g., from text to text) and across different interfaces (e.g., from text to image or from text to video). Well-known generative AI tools include Bard, DALL-E, Midjourney, ChatGPT, and Deepmind.

Generative AI counts on neural network techniques such as transformers, GANs (generative adversarial networks), and VAEs for problem-solving. In contrast, conventional AI algorithms often adhere to a predefined set of rules to process data and generate results

using convolutional neural networks, recurrent neural networks, and reinforcement learning. An issue with conventional AI is that algorithms can magnify or reproduce pre-existing biases and prejudices present in training data. A notable example is Amazon's AI-powered recruiting tool, abandoned due to gender bias.

Generative AI finds extensive applications across commercial domains, automatically generating new material and helping to interpret and understand already-existing content. Virtually any kind of content can be generated by employing it. Some of the use cases of generative AI include:

- Writing emails, resumes, dating profiles, and term papers
 - Simulating realistic human motions used in gaming, animation and virtual reality
 - Assisting in the discovery of new drugs by recommending potential chemical structures
 - Generating synthetic climate data to train a model that can predict weather
- Generative AI developers are focusing on enhancing user interaction, allowing them to ask questions in plain language. After receiving an initial response, the results can be customised by suggesting the tone, style, and any other elements users want included in the created material.